Introduction

Seasonal ARIMA (SARIMA) extends the standard ARIMA framework to series with periodic seasonality. The standard notation is ARIMA$(p, d, q)(P, D, Q)_s$ where the lower-case letters are non-seasonal orders, the upper-case are seasonal orders, and s is the seasonal period (12 for monthly with annual seasonality, 4 for quarterly, 24 for hourly with daily seasonality). SARIMA is the workhorse parametric forecasting tool for economic, financial, and many biomedical time series with stable seasonal patterns.

Prerequisites

A working understanding of ARIMA models, differencing, the backshift operator, and seasonal decomposition.

Theory

The full SARIMA equation is

$$\Phi_P(B^s)\phi_p(B)(1-B)^d(1-B^s)^D y_t = \Theta_Q(B^s)\theta_q(B)\varepsilon_t,$$

where B is the backshift operator ($By_t = y_{t-1}$). The non-seasonal AR, differencing, and MA components compose multiplicatively with their seasonal counterparts at lag s. The model accommodates trends through differencing (d) and seasonal patterns through both seasonal differencing (D) and seasonal AR/MA terms.

The famous "airline model" $(0, 1, 1)(0, 1, 1)_{12}$ for monthly airline passengers — Box and Jenkins's headline example — demonstrates that even highly trending and strongly seasonal series can fit into the SARIMA family with modest order.

Assumptions

The series is stationary after regular and seasonal differencing; residuals are white noise; the seasonal period s is correctly specified by the `frequency()` of the input `ts` object.

R Implementation

```r
library(forecast)

x <- AirPassengers

fit <- auto.arima(log(x), seasonal = TRUE)
summary(fit)

fc <- forecast(fit, h = 24)
autoplot(fc)

checkresiduals(fit)
```

Output & Results

`auto.arima()` searches over candidate orders using AICc and returns the chosen SARIMA model with its coefficients and standard errors. `forecast()` produces point forecasts and 80 % / 95 % prediction intervals. `checkresiduals()` reports the Ljung-Box test, ACF, and time-series plot of residuals.

Interpretation

A reporting sentence: "`auto.arima` on log-transformed AirPassengers selected an ARIMA$(0, 1, 1)(0, 1, 1)_{12}$ structure (AICc $= -483$, $\sigma^2 = 0.0014$); residuals passed Ljung-Box at lag 24 ($p = 0.61$). Forecasts for the next 24 months show continued multiplicative growth and stable seasonal pattern, with 95 % prediction intervals widening from ±5% at 1 month to ±18% at 24 months." Always check residual diagnostics before reporting forecasts.

Practical Tips

- Use `auto.arima()` for automatic order selection; double-check the result against ACF/PACF inspection of the differenced series.
- Log-transform for multiplicative seasonality (where seasonal amplitude grows with the level); SARIMA on the log scale is then additive and well-behaved.
- For very long seasonal periods ($s \geq 52$), SARIMA becomes computationally heavy; consider regression with Fourier seasonal terms (`forecast::tslm` with `fourier()`) instead.
- Always use chronological train/test splits; random CV is invalid for time series because it leaks future information into past folds.
- Box-Jenkins residual diagnostics (Ljung-Box, ACF, PACF, normality plot) are essential after fitting; `checkresiduals()` aggregates them.
- For multiple seasonalities (weekly + yearly), use TBATS (`forecast::tbats`) or `mstl()` decomposition with non-seasonal forecasting on the adjusted series.

R Packages Used

`forecast::auto.arima()` and `Arima()` for canonical SARIMA fitting and forecasting; `fable::ARIMA()` for the modern tidyverts interface; `feasts::ACF()` and `PACF()` for diagnostic visualisations; `seasonal::seas()` for X-13ARIMA-SEATS as the official-statistics alternative.

For Reviewers

What to look for in a paper using this method.

- **Common misapplications.**
- **Diagnostics that should be reported but often aren't.**
- **Red flags in tables and figures.**
- **What to verify.**
- **What an adequate Methods paragraph must contain.**

See also — labs in this chapter

- Time series basics